

Quiz (Carolina Reaper)

Q1. Factor $x^2 - 4$.

- (A) $x^2 - 2^2$
- (B) $(x - 2)(x + 2)$
- (C) $x(x - 2)$
- (D) $(x - 2)(x - 2)$
- (E) $x^2 - 2x + 4$

Q2. Factor $25 - y^2$.

- (A) $(5 - y)(5 + y)$
- (B) $5(5 - y)$
- (C) $(5 + y)(5 - y)$
- (D) $25 - y$
- (E) $5(5 + y)$

Q3. Factor $x^2 - 9y^2$.

- (A) $(x - 3y)(x + 3y)$
- (B) $x(x - 3y)$
- (C) $(x - 3y)(x - 3y)$
- (D) $(x + 3y)(x - 3y)$
- (E) $x^2 - 3y$

Q4. Simplify $(a^2 - b^2) + (b^2 - c^2)$.

- (A) $a^2 - c^2$
- (B) $b^2 - c^2$
- (C) $a^2 + b^2 - c^2$
- (D) $a^2 - b^2 + c^2$
- (E) $a^2 + c^2$

Q5. Factor $36 - 4x^2$.

- (A) $2(3 - x)(3 + x)$
- (B) $2(3 + x)(3 + x)$
- (C) $4(3 - x)(3 + x)$
- (D) $6(6 - 2x)$
- (E) $2(3 - x)(3 - x)$

Q6. Solve $(x - 2)(x + 2) = 0$.

- (A) $x = 0, 2$
- (B) $x = 2, -2$
- (C) $x = 4, -4$
- (D) $x = 1, -1$
- (E) $x = 3, -3$

Q7. Factor $4x^2 - 9$.

- (A) $3(3 - 2x)$
- (B) $(2x - 3)(2x + 3)$
- (C) $(3 - 2x)(3 + 2x)$
- (D) $2x(2x - 3)$
- (E) $2x(2x + 3)$

Q8. Simplify $(x^2 - y^2) - (x^2 + y^2)$.

- (A) $2x^2 - 2y^2$
- (B) $-2y^2$
- (C) $2y^2$
- (D) $-2x^2$
- (E) $2x^2 + 2y^2$

Open Ended Questions (FRQ)

Q9. Factor $x^4 - y^4$ using difference of squares twice and explain your steps.

Q10. Use difference of squares to simplify the expression $(a + b)(a - b) + (c + d)(c - d)$ and show your work.

Chef's Tips

- Remind students to check their signs when factoring.
- Encourage students to simplify expressions after factoring.
- Emphasize the importance of writing the final answer in the correct format.